

Evidence-Grounded Quarterly Performance Reporting

Architecture and the AI boundary

The system takes client code and quarter, discovers sources by content, and builds one report model for PDF and evidence review. Python owns extraction, calculations, chronology, fund selection, ranking and compliance. PyMuPDF, openpyxl, ReportLab and Matplotlib keep the runtime portable without Office or downloaded document models. A one-command launcher installs pinned dependencies.

One bounded evidence agent chooses read-only tools: list, scoped search, inspect a registered passage, validate an exact quote, finish or escalate. Each action receives validation feedback before the next decision. It has an eight-action limit and repeated-action stop; it cannot alter numbers or approve reports. Its validated focus quotes feed narrative synthesis: a matching fund narrative must cite at least one focus quote or be rejected. This is a hybrid agentic workflow, not an autonomous investment analyst. A shared retrieval service could replace the local passage registry without a separate vector database.

Verification and regression detection

Models read chart images and write source-grounded connecting prose. Chart schema checks labels, units, duplicates, finite values and axis bounds, with one repair; gridline readings remain approximate and require confirmation. Python computes cap-rate spread direction. Writing uses source-scoped numerical placeholders, including occupancy and asset names. Required slots appear exactly once; code inserts exact strings and rejects raw/cross-source numbers and changed units. The host attaches exact quotes to cited sources; narrower model quotes are checked verbatim. A separate model checks paraphrase meaning before one bounded repair. Neither judgment nor quote presence proves entailment; human review remains required. Rejected prose retains complete attributed excerpts, never sentence trimming. Market synthesis covers house views, risks, material exposures, forward NOI, broad returns and valuation risk; other charts stay in the audit. OpenAI defaults to pinned GPT-4.1 prose and separately evaluated GPT-4.1-mini vision.

Money arithmetic uses Decimal; currency identities have zero tolerance. Fractions of a cent, booleans and nonfinite values are rejected; JSON preserves base-ten strings. Display rounding never overwrites sources. Contribution is income plus appreciation less fees; flows belong only in roll-forwards. Rankings retain overlapping roles; TWR/IRR and weighted LTV are read. Rounded manager statistics are precision-aware corroboration. IC/one-pager amounts match exactly; history uses displayed USD-million precision, with missing client ownership disclosed.

Golden/adversarial tests cover one-cent errors, shifted/renamed inputs, added holdings, duplicates, invalid IC rows, fabricated quotes, agent limits, chart failures and approval tampering. Ten fixed-seed randomized cases permute discovery/IC inputs or corrupt cash values, not unrestricted fuzzing. The regression matrix records outcomes. Independent pairs of no-key and live API runs with empty caches passed canonical comparison: financial/chart figures, fund selection/order/role, compliance and structure. Full live payloads differed; byte-identical prose is not claimed. Temperature zero, snapshots, seed and caching help but do not guarantee provider determinism; semantic differences fail the test.

Human review and finalization

Clicking report blocks opens locators and recursive calculation inputs. The review queue distinguishes warnings, blockers and required chart confirmations. Meridian's missing report permits flash-only commentary, never invented asset drivers. Missing policy, ambiguous required sources, currency reconciliation failures and unavailable chart readings prevent finalization. Existing automatic chart readings can be corrected, but each correction must pass the same schema and include a reviewer reason. Money corrections belong in the authoritative workbook followed by regeneration.

Approval requires a named acknowledgment, chart confirmations, no blockers, and unchanged draft/source/asset hashes. It creates a final PDF and hash-bound approval/audit records. Regeneration invalidates approval. This guards accidental stale release, not a hostile filesystem owner. No-key mode retains deterministic finance, source excerpts and report sections, but disables the agent and image readings. The supplied live draft still requires human review/approval.

Next quarter and format changes

A full-pipeline rehearsal generates a different client, 1Q26 sources, renamed files, moved rows, one removed and two added holdings, and an approval carried from a 2025 log into a 2026 withdrawal. It completes without production edits; its 13 holdings, NAV and surviving approval match fixture expectations. This no-key rehearsal tests adaptation, not live vision accuracy. Discovery records accepted/ignored/failed files and reasons. Tables use labels/grouped headers, not coordinates. Canonical entity rules normalize punctuation, legal suffixes and Roman numerals; collisions and ambiguous matches are not guessed.

Client identity and benchmark come from the flash; policy comes from the prior PMR. Entity rules cover word/Roman/Arabic vintages, ampersands, legal suffixes and explicit abbreviations. Unmatched documents receive suggestions for review, never automatic fuzzy merges or false absence claims. The prior PDF supplies typography, color and contents order. Blocks flow as holdings grow. Unsupported policy syntax blocks release; new structure is flagged. Arbitrary redesign requires an adapter, not universal document understanding.

Specific judgments and remaining work

Traceability is a machine-readable claim manifest plus fact ledger: file, sheet/cell, page or slide/chart, raw values, numeric metadata, formulas and input IDs. Attribution cells have individual references; other blocks have scoped evidence references. Run diagnostics expose stage durations, calls/usage, cache hits/misses, failures and the agent timeline without hidden reasoning or credentials. All outputs remain inspectable without a monitoring service.

The SPEC-designated FundingStatus/AnnReturns rows own portfolio summary/compliance; Vectera Initiated Investments owns explicitly labeled funded-program inception statistics. The prior attribution table omits a selected individual position: this table includes all required roles. The sample's 'Of which funded' figure is a commitment amount; this report labels it 'Flash commitment total.' The realized-only \$0.51 discrepancy is outside the active portfolio and is a warning. Broad market returns are not conflated with the client benchmark; mandate appointment dates and performance inception are not assumed to mean the same thing. The appendix retypesets parsed PDF tables on portrait Letter with single-row headers and source-page captions, preserving displayed values and blank cells; the original six-page PDF is retained.

Live OpenAI access has been observed on macOS/Python 3.12. Windows 11 bootstrap/interactive execution and genuine final human approval remain outstanding, not claimed as verified. Unknown manager headings retain page evidence and raise an extraction warning, never a false missing-report claim. Attribution displays rounded whole dollars and accounting negatives; exact cents remain authoritative in the audit. Negative fees are preserved as credits and flagged. With more time: authenticated approvals, broader scanned-document/policy adapters, deterministic chart measurement and more cross-client fixtures. Terraform, Docker dependence and multiple agents were deliberately omitted.